

Lesson 3-3 Quiz: Polynomial Identities

Name: _____

Period: _____

Question 1: Polynomial Identities (Factoring)

/6 pts

Use a polynomial identity to factor $8 + 125x^3$.

(a) Which polynomial identity applies here? Circle one.

(1 pt)

Difference of Cubes	Sum of Cubes
$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$	$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

(b) Show your work. Factor $8 + 125x^3$ completely.

(3 pts)

Word Bank						
2	$5x$	4	$25x^2$	$10x$	sum of cubes	difference of cubes

Identify a and b : $a = \underline{\hspace{2cm}}$ $b = \underline{\hspace{2cm}}$

Substitute into the identity:

Final factored form: $8 + 125x^3 = (\underline{\hspace{2cm}})(\underline{\hspace{2cm}})$

(c) In 1–2 sentences, explain how you knew which identity to use.

(2 pts)

Question 2: Binomial Theorem Expansion

/4 pts

Use the Binomial Theorem to expand $(x - 3)^5$.

Word Bank						
Pascal's Triangle coefficients for $n = 5$: 1 5 10 10 5 1						

(a) Fill in the missing coefficients from Pascal's Triangle:

(1 pt)

$$(x - 3)^5 = \underline{\hspace{1cm}}x^5(-3)^0 + \underline{\hspace{1cm}}x^4(-3)^1 + \underline{\hspace{1cm}}x^3(-3)^2 \\ + \underline{\hspace{1cm}}x^2(-3)^3 + \underline{\hspace{1cm}}x^1(-3)^4 + \underline{\hspace{1cm}}(-3)^5$$

(b) Evaluate each power of (-3) and simplify each term:

(2 pts)

(c) Write the expanded polynomial in standard form:

(1 pt)

Final Answer: $(x - 3)^5 = \underline{\hspace{4cm}}$

Total: /10

Lesson 3-3 Quiz — Scoring Rubric

Teacher Copy · 10 points total

Question 1: Polynomial Identity Factoring

6 points

Part	Criteria	Points	Answer Key
(a)	Correctly identifies Sum of Cubes	1 pt	Sum of Cubes
(b)	Correctly identifies $a = 2$ and $b = 5x$	1 pt	$a = 2, b = 5x$
(b)	Shows substitution into identity	1 pt	$(2 + 5x)(4 - 10x + 25x^2)$
(b)	Correct final factored form	1 pt	$(2 + 5x)(4 - 10x + 25x^2)$
(c)	States the expression is a <i>sum</i> (uses +)	1 pt	<i>See sample below</i>
(c)	Identifies both terms as perfect cubes ($2^3 = 8, (5x)^3 = 125x^3$)	1 pt	<i>See sample below</i>

Sample response for (c)

"I knew to use the sum of cubes identity because the expression uses addition (+), and both 8 and $125x^3$ are perfect cubes: $2^3 = 8$ and $(5x)^3 = 125x^3$."

Question 2: Binomial Theorem Expansion

4 points

Part	Criteria	Points	Answer Key
(a)	Fills in all 6 Pascal's Triangle coefficients correctly: 1, 5, 10, 10, 5, 1	1 pt	Coefficients provided in word bank
(b)	Correctly evaluates powers of (-3) : 1, -3 , 9, -27 , 81, -243	1 pt	Common error: all positive (ignoring negative sign)
(b)	Correctly multiplies coefficients $\times$ powers of (-3) to simplify each term	1 pt	1, -15 , 90, -270 , 405, -243
(c)	Writes complete polynomial in standard form	1 pt	See key below

Answer Key

$$\begin{aligned}(x - 3)^5 &= \binom{5}{0}x^5(-3)^0 + \binom{5}{1}x^4(-3)^1 + \binom{5}{2}x^3(-3)^2 + \binom{5}{3}x^2(-3)^3 + \binom{5}{4}x^1(-3)^4 + \binom{5}{5}x^0(-3)^5 \\ &= x^5 - 15x^4 + 90x^3 - 270x^2 + 405x - 243\end{aligned}$$

Topic 3 Assessment Alignment:

Q1 (Sum of Cubes) → Topic Assessment Q8: Factor $64 + 27a^3$

Q2 (Binomial Expansion) → Topic Assessment Q3: Expand $(x - 2)^6$ and Q13: Expand $(2a + 2b)^5$

Lesson 3-3 Quiz — Sentence Frames

ELL Support Version · Use with Question 1, Part (c)

Question 1(c): In 1–2 sentences, explain how you knew which identity to use.

Option A — Fill in the Blanks

I used the _____ identity because the two terms are connected by _____.

I know that _____ is a perfect cube because _____³ = _____,

and _____ is a perfect cube because _____³ = _____.

Word Bank

8 $125x^3$ 2 $5x$ addition (+) subtraction (–)
sum of cubes difference of cubes

Option B — Sentence Starters

I chose the _____ identity because the expression $8 + 125x^3$ shows _____.

Both terms are perfect cubes because _____

Key Vocabulary

identity	A math equation that is always true for all values of the variable.
perfect cube	A number or expression that can be written as something raised to the 3rd power. <i>Examples:</i> $8 = 2^3$ $125x^3 = (5x)^3$ $27 = 3^3$
sum	The result of adding . The symbol + means sum.
difference	The result of subtracting . The symbol – means difference.
factor	To rewrite an expression as a product (multiplication) of simpler parts.